IBM COGNOS ANALYTICS
PROJECT REPORT

Juvenile Justice Disposal

Analysis Report — 2023

Disposal of Juveniles Apprehended under IPC & SLL | 19 Major Indian Cities

Dataset	NCRB Crime in India 2023 — Table 5B-5
Tool	IBM Cognos Analytics — Report Studio
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Academic Year	2026–2027
Repository	https://github.com/Vikas-Yadav-6696/juvenile-justice-cognos-analytics-india

1. Executive Summary

This report presents a comprehensive analysis of NCRB's Crime in India 2023, Table 5B-5, covering the disposal of juveniles apprehended under the Indian Penal Code (IPC) and Special & Local Laws (SLL) across 19 major Indian cities during the calendar year 2023.

The dataset captures 16,200 total juvenile cases — comprising 7,521 fresh apprehensions and 8,679 cases carried forward from 2022. A critical system-level concern is the overall case resolution rate of only 43.9%, meaning 9,086 cases remained pending at year-end. The average guilt determination rate stands at 92.5%, reflecting the predominance of rehabilitative disposals in India's juvenile justice framework.

Performance Snapshot — 2023

TOTAL CASES	FRESH APPREHENSIONS	PENDING DISPOSAL	AVG. GUILTY VERDICT
16,200	**7,521**	**9,086**	**92.5%**
Incl. 8,679 carried forward	New cases in 2023	56.1% backlog rate	Rehabilitation-led

Table 1: National KPI summary. Red indicates critical concern; Green indicates positive performance.

1.1 Scope & Coverage

The analysis covers 19 Commissionerate/City Police units spanning metro cities (Delhi, Mumbai, Chennai, Kolkata, Bengaluru, Hyderabad) and Tier-2 cities (Indore, Jaipur, Surat, Nagpur, etc.). One city — Ghaziabad — reported zero activity across all fields, suggesting a data reporting gap rather than genuine absence of juvenile cases.

2. Dataset Overview & Methodology

2.1 Data Source

Source: National Crime Records Bureau (NCRB), Ministry of Home Affairs, Government of India.

Publication: Crime in India 2023 — Table 5B-5: Details of Disposal of Juveniles Apprehended under IPC & SLL.

Reporting Period: January 1, 2023 – December 31, 2023.

2.2 Column Definitions

1. Pending at Beginning (Col.3): Cases carried forward from the previous year.
2. Apprehended During Year (Col.4): New juvenile cases registered in 2023.
3. Total (Col.5): Sum of pending + apprehended — total cases on the system in 2023.
4. Discharged During Investigation (Col.6): Cases closed as unfounded or quashed.
5. Sent Home After Advice/Admonition (Col.7): Non-punitive restorative disposal — core to JJ Act 2015.
6. Sent to Special Home/Fit Institution (Col.8): Institutionalised rehabilitation placement.

7. Dealt with Fine (Col.9): Punitive monetary disposal.
8. Awarded Imprisonment (Col.10): Custodial sentence — should be last resort under JJ Act 2015.
9. Acquitted or Discharged (Col.11): Cases resulting in formal acquittal.
10. Percentage Held Guilty (Col.12): Proportion of disposed cases with a guilty finding.
11. Pending Disposal (Col.13): Cases still unresolved at December 31, 2023.

3. Primary Data Table — Juvenile Disposal by City (2023)

The table below presents original NCRB data with one computed column — Resolution Rate — added for immediate performance visibility. Colour coding: Green (≥75%), Amber (30–74%), Red (<30%).

City	Pending (Start)	Apprehended	Total Cases	Sent Home	Special Home	Imprisoned	Pending (End)	Guilty %	Resolution Rate
Ahmedabad	0	460	460	414	46	0	0	100%	**100%**
Bengaluru	97	**551**	648	287	142	0	176	100%	**72.8%**
Chennai	1337	**771**	2108	0	10	**102**	**1651**	37.5%	**21.7%**
Coimbatore	107	37	144	21	0	0	99	57.8%	**31.3%**
Delhi	1698	**3098**	4796	1649	895	**1**	**1575**	94.6%	**67.2%**
Ghaziabad	0	0	0	0	0	0	0	N/A	**N/A**
Hyderabad	25	241	266	198	52	0	16	100%	**94%**
Indore	482	237	719	47	156	**42**	233	86.2%	**67.6%**
Jaipur	168	380	548	103	327	0	117	100%	**78.6%**
Kanpur	103	48	151	0	0	0	151	N/A	**0%**
Kochi	0	56	56	54	1	0	1	100%	**98.2%**
Kolkata	0	133	133	80	10	0	0	73.1%	**100%**
Kozhikode	46	58	104	49	0	0	50	90.7%	**51.9%**
Lucknow	37	0	37	0	0	0	37	N/A	**0%**

City	Pending (Start)	Apprehended	Total Cases	Sent Home	Special Home	Imprisoned	Pending (End)	Guilty %	Resolution Rate
Mumbai	1738	342	2080	109	0	0	**1964**	94%	**5.6%**
Nagpur	229	373	602	476	0	0	116	99.8%	**80.7%**
Patna	249	50	299	0	0	0	299	N/A	**0%**
Pune	2363	443	2806	194	0	0	**2601**	100%	**7.3%**
Surat	0	243	243	0	243	0	0	100%	**100%**
TOTAL — 19 Cities	**8,679**	**7,521**	**16,200**	**3,681**	**1,882**	**145**	**9,086**	**92.5%**	**43.9%**

Table 2: Primary disposal statistics across 19 Indian cities. Source: NCRB CII 2023, Table 5B-5.

4. Derived Analytical Columns

Five specialised metrics were computed from the source data to enable evidence-based benchmarking, resource planning, and JJ Act 2015 compliance tracking. These columns go beyond what NCRB publishes and form the analytical core of this IBM Cognos project.

4.1 Metric Definitions & Formulae

1 Resolution Rate %

= (Total Cases − Pending End) ÷ Total Cases × 100

Measures the proportion of all cases (new + inherited) disposed during the year. A rate below 50% means the system is accumulating more caseload than it clears. National figure: 43.9%.

2 Rehabilitation Rate %

= (Sent Home + Sent to Special Home) ÷ Total Cases × 100

Captures the share receiving non-punitive, restorative disposals — the primary intent of the Juvenile Justice (Care and Protection of Children) Act, 2015.

3 Conviction Rate %

= (Fined + Imprisoned) ÷ Total Cases × 100

Tracks purely punitive outcomes. Should ideally be as low as possible in a rehabilitative justice framework. Indore (30.04%) and Chennai (5.27%) are outliers.

4 Backlog Ratio

= Cases Pending at Beginning ÷ Apprehended During the Year

A ratio >1 indicates inherited backlog exceeds current-year apprehensions — a systemic judicial capacity stress signal. Mumbai (5.08) and Pune (5.33) are critical.

5 Severity Index %

= Juveniles Awarded Imprisonment ÷ Total Cases × 100

Key human-rights indicator. Any city consistently above 3% warrants a judicial conduct review. Nationally only 0.89% — largely JJ Act compliant.

4.2 Derived Metrics — Complete City Table

Risk Level is derived from the Pending Disposal share: HIGH (≥70% of total pending), MEDIUM (40–69%), LOW (<40%).

City	Resolution Rate %	Rehab. Rate %	Conviction Rate %	Backlog Ratio	Severity Index %	Risk Level
Ahmedabad	**100%**	100%	0%	0	0%	**LOW**
Bengaluru	**72.8%**	66.2%	0%	0.18	0%	**LOW**
Chennai	**21.7%**	0.5%	4.84%	1.73	4.84%	**HIGH**
Coimbatore	**31.3%**	14.6%	0%	**2.89**	0%	**MEDIUM**
Delhi	**67.2%**	53%	0.02%	0.55	0.02%	**LOW**
Ghaziabad	**N/A**	N/A	N/A	N/A	N/A	**N/A**
Hyderabad	**94%**	94%	0%	0.1	0%	**LOW**
Indore	**67.6%**	28.2%	5.84%	**2.03**	5.84%	**LOW**
Jaipur	**78.6%**	78.5%	0%	0.44	0%	**LOW**
Kanpur	**0%**	0%	0%	**2.15**	0%	**HIGH**
Kochi	**98.2%**	98.2%	0%	0	0%	**LOW**
Kolkata	**100%**	67.7%	0%	0	0%	**LOW**
Kozhikode	**51.9%**	47.1%	0%	0.79	0%	**MEDIUM**
Lucknow	**0%**	0%	0%	N/A	0%	**HIGH**
Mumbai	**5.6%**	5.2%	0%	**5.08**	0%	**HIGH**
Nagpur	**80.7%**	79.1%	0%	0.61	0%	**LOW**
Patna	**0%**	0%	0%	**4.98**	0%	**HIGH**
Pune	**7.3%**	6.9%	0%	**5.33**	0%	**HIGH**
Surat	**100%**	100%	0%	0	0%	**LOW**
TOTAL (All Cities)	**43.9%**	**34.2%**	**2.50%**	**1.54**	**0.89%**	**MEDIUM**

Table 3: All five computed metrics per city. Colour-coded values reflect performance against national benchmarks.

5. Key Findings

Finding 1 — Maharashtra Backlog Crisis

Mumbai (Backlog Ratio: 5.08) and Pune (5.33) carry inherited caseloads more than 5× larger than their annual apprehensions, indicating a structural judicial capacity gap. Without intervention, these backlogs will continue to compound year-on-year.

Finding 2 — Exemplary Disposal Efficiency

Ahmedabad, Kolkata, and Surat achieved 100% annual resolution, completely clearing their year-start backlogs. Kochi reached 98.2% and Hyderabad 94.0%, demonstrating that high resolution rates are practically achievable.

Finding 3 — Rehabilitation Dominance (JJ Act Aligned)

Nationally, 34.4% of cases resulted in juveniles being sent home with advice and 11.6% placed in Special Homes — a combined rehabilitation rate of 46.1%, aligned with the JJ Act 2015 mandate to treat custodial sentencing as a last resort.

Finding 4 — Imprisonment Concerns in Two Cities

Chennai recorded a Severity Index of 4.84% (102 juveniles imprisoned) and Indore 5.84% (42 imprisoned). Both cities significantly exceed the national average of 0.89% and require immediate judicial review.

Finding 5 — Inactive/Incomplete Reporting Jurisdictions

Ghaziabad reported zero activity across all fields — likely a reporting failure rather than genuine absence of juvenile cases. Lucknow (37 pending, zero apprehensions) and Kanpur (151 pending, 48 apprehended but no disposals) also show data completeness gaps.

Finding 6 — Delhi Dominates National Volume

Delhi accounts for 41.2% of all fresh apprehensions across the 19 cities (3,098 of 7,521), making it the single most resource-intensive juvenile justice jurisdiction in India. Its 67.2% resolution rate, while above average, still leaves 1,575 cases pending.

6. Notable City Profiles

6.1 Delhi City — Volume Leader

Delhi is the juvenile justice system's most resource-intensive jurisdiction with 4,796 total cases and 3,098 new apprehensions in 2023. It achieved a 67.2% resolution rate, placing 1,649 juveniles back home and 895 in Special Homes — the highest Special Home placements in the dataset. The Backlog Ratio of 0.55 is manageable. Guilty verdict rate: 94.6%.

6.2 Mumbai — Acute Backlog

Mumbai entered 2023 with 1,738 cases pending. Despite receiving only 342 new apprehensions, just 116 cases were disposed. The pending caseload grew to 1,964 — a 13% increase in the backlog burden. Its Backlog Ratio of 5.08 is the second-highest nationally and demands immediate judicial capacity intervention.

6.3 Pune — Most Critical Backlog

Pune presents the most acute backlog situation nationally. It entered 2023 with 2,363 pending cases, received 443 new apprehensions, yet disposed of only 205 cases (resolution rate: 7.3%). The pending caseload grew to 2,601 — a 10.1% worsening — giving it the highest Backlog Ratio of 5.33.

6.4 Indore — Highest Conviction Rate

Indore is a unique and concerning outlier with a Conviction Rate of 30.04% — the highest nationally by a large margin. Of 719 total cases, 174 juveniles were dealt fines and 42 were imprisoned, yielding a Severity Index of 5.84%. Both figures require a formal judicial conduct review in the context of JJ Act 2015 compliance.

6.5 Best-Practice Cities (Bengaluru, Jaipur, Hyderabad, Kochi)

These four cities consistently demonstrate best-practice profiles: Bengaluru (72.8% resolution, zero imprisonments), Jaipur (78.6%, 100% rehabilitation rate), Hyderabad (94% resolution, zero imprisonments, 94% rehabilitation), and Kochi (98.2%, 98.2% rehabilitation, zero imprisonments). These cities provide a proven operational template for underperforming jurisdictions.

7. Policy Recommendations

The following recommendations are evidence-based, derived directly from the data analysis and cross-referenced against the Juvenile Justice (Care and Protection of Children) Act, 2015. They are ranked by urgency and potential impact.

No.	Focus Area	Key Cities	Recommendation	Priority	
1	**Backlog Reduction**	Mumbai, Pune, Chennai, Patna	Establish dedicated fast-track Juvenile Justice Boards (JJBs) in Maharashtra. Set mandatory 90-day case disposal targets with monthly performance dashboards reported to the state Home Ministry.	**HIGH**	
2	**Reduce Imprisonment**	Chennai, Indore	Align sentencing with JJ Act 2015 — custodial sentences must be the absolute last resort. Mandate judicial training on restorative justice and pre-sentencing rehabilitation assessments for every juvenile.	**HIGH**	
3	**Rehabilitation Infrastructure**	Indore, Coimbatore, Kozhikode	Expand community-based rehabilitation: open shelters, counselling centres, and skill development workshops. Reduce fines and Special Home placements for minor offences. Target rehabilitation rate of 65%+ nationally.	**HIGH**	
4	**Data Quality & Reporting**	Ghaziabad, Lucknow, Kanpur	Mandate monthly case-level data uploads to NCRB via a standardised digital portal. Trigger automatic administrative audits for jurisdictions filing zero or incomplete records.	**MEDIUM**	
5	**Special Home Oversight**	Delhi, Jaipur, Surat	Audit placement standards, capacity, and exit outcomes of all Special Homes/Fit Institutions. Delhi (895 placements) and Jaipur (327) require immediate inspection for overcrowding.	**MEDIUM**	
6	**Scale Best Practices**	Ahmedabad, Kochi, Kolkata	Document and publish disposal frameworks from cities achieving 100% resolution. Create a national knowledge exchange programme for laggard jurisdictions to adopt proven models by 2027.	**LOW**	

Table 4: Policy recommendations with priority ratings. HIGH = immediate action required; MEDIUM = short-term planning; LOW = long-term strategy.

7.1 Implementation Roadmap

Immediate (0–3 months)

- Fast-track JJB setup in Mumbai and Pune.
- Judicial training on JJ Act 2015 for Chennai and Indore benches.
- Emergency audit of Special Homes in Delhi for overcrowding.

Short-term (3–12 months)

- Launch national digital NCRB reporting portal for city-level data.
- Standardise rehabilitation programme evaluation metrics across states.
- Constitute state-level Backlog Reduction Task Forces in Maharashtra and Tamil Nadu.

Long-term (1–3 years)

- National knowledge exchange: top-performing cities mentoring laggards.
- Target a national resolution rate of 70%+ by 2027.
- Establish a Juvenile Justice Performance Index (JJPI) tracked and published annually by NCRB.

8. Conclusion

This analysis of NCRB 2023 juvenile justice data across 19 Indian cities reveals a system simultaneously exhibiting pockets of excellence and zones of systemic distress. Cities like Ahmedabad, Kochi, and Surat demonstrate that complete annual resolution is achievable. Maharashtra's metro cities — Mumbai and Pune — represent a national crisis in juvenile case backlog that demands urgent judicial and administrative reform.

The five derived analytical metrics introduced in this report — Resolution Rate, Rehabilitation Rate, Conviction Rate, Backlog Ratio, and Severity Index — provide a multidimensional lens beyond the raw NCRB figures. Together, they enable evidence-based benchmarking, targeted resource allocation, and compliance monitoring with the Juvenile Justice (Care and Protection of Children) Act, 2015.

IBM Cognos Analytics, through its reporting and visualisation capabilities, provides an ideal platform for deploying these derived metrics as live, interactive dashboards — enabling real-time monitoring by state government agencies, the National Commission for Protection of Child Rights (NCPCR), and district-level Juvenile Justice Boards nationwide.

Key Takeaway: With a national resolution rate of only 43.9% and 9,086 cases pending at year-end, the Indian juvenile justice system faces a structural backlog crisis. Targeted interventions in Maharashtra and data-driven governance using IBM Cognos Analytics can close this gap within 3 years.

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